

Module 2

Transient Analysis of First and Second-Order Circuits

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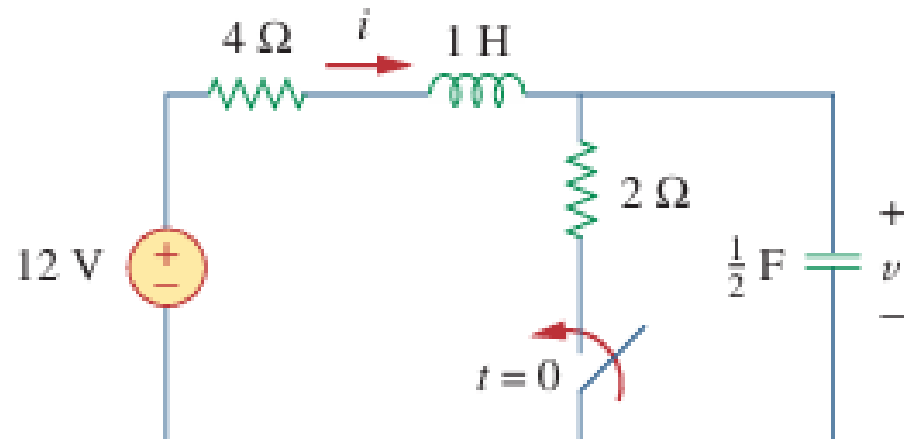
General Second Order circuits

1. We first determine the initial conditions and the final value.
2. We turn off the independent sources and find the form of the transient response by applying KCL and KVL. Once a second-order differential equation is obtained, we determine its characteristic roots. Depending on whether the response is overdamped, critically damped, or underdamped, we obtain with two unknown constants.
3. We obtain the steady-state response.
4. The total response is now found as the sum of the transient response and steady-state response
5. We finally determine the constants associated with the transient response by imposing the initial conditions and determined in step 1.

We can apply this general procedure to find the step response of any second-order circuit, including those with op amps.

Exercise

Find the complete response v and then I for $t > 0$.



Next class: Module 3